

SMART INDIA HACKATHON 2026
IDEA SUBMISSION — DETAILED PROJECT REPORT

FLEETGRID

Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots in Smart Warehouses

Problem Statement ID: 26123

Theme: Smart Automation

PS Category: Software

Team Name: Vector6

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1. Executive Summary

FleetGrid is a serverless, edge-AI coordination layer for autonomous mobile robot (AMR) fleets operating in smart warehouses. Instead of routing every navigation and task-allocation decision through a central fleet-management server, each robot in FleetGrid acts as an intelligent, self-sufficient node: it senses its surroundings, exchanges intent with nearby robots over a peer-to-peer mesh, bids for tasks and right-of-way using local rules, and re-plans instantly when conflicts arise.

The system is built around three cooperating mechanisms — a decentralized Gossip Protocol for task management, Market-Based Intersection Bidding (MBIB) for instant local conflict resolution at choke points, and Spatial Broadcast Limiting for scalable, spam-free mesh communication — all running on commercial off-the-shelf (COTS) edge hardware (NVIDIA Jetson Nano / Raspberry Pi 4) over a BATMAN-adv mobile ad-hoc mesh network.

This report consolidates the team's SIH idea submission, a working simulation prototype (“FleetGrid” digital-twin dashboard), and the academic grounding behind decentralized multi-robot coordination — anchored by Li, Gama, Ribeiro & Prorok (2020), “Graph Neural Networks for Decentralized Multi-Robot Path Planning” — into a single, editable project document.

2. Problem Statement

Problem Statement Title: Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots in Smart Warehouses.

Modern warehouse AMR fleets are typically coordinated through a central fleet-management server. As deployments scale, this architecture creates four recurring failure modes:

- Wi-Fi drops create communication gaps — a single wireless dead zone can silently strand robots mid-task.
- Central servers risk downtime and vulnerability — one failure point can halt the entire fleet.
- “Stop-and-wait” behaviour slows fleet operations — robots idle at intersections awaiting centralized arbitration.
- Dependence on a master controller limits autonomy — fleets cannot scale or self-heal without upgrading central infrastructure.

These constraints become more expensive and more fragile exactly as a warehouse tries to scale up automation — the opposite of what a growing operation needs.

3. Proposed Solution & Innovation

FleetGrid replaces the central controller with a serverless, peer-to-peer AMR swarm in which every robot is an intelligent node that senses, decides, and communicates locally.

3.1 Core Mechanisms

Market-Based Intersection Bidding (MBIB)

Autonomous robots resolve choke-point conflicts instantly by running local bids based on payload priority, battery level, and distance to goal. This eliminates server round-trip delays and keeps fleet movement fluid.

Decentralized Gossip Protocol

Tasks are broadcast and managed via a gossip protocol rather than a central dispatcher, removing reliance on a single server for task assignment.

BATMAN-adv Mobile Ad-hoc Mesh

AMRs route data through each other using a BATMAN-adv mesh network, so the fleet's communication fabric stays alive even if individual links or nodes drop out.

Spatial Broadcast Limiting

Robots broadcast telemetry only within a 5-metre radius using spatial hashing. This prevents peer-to-peer message spam and ensures the mesh scales smoothly as more units are added.

3.2 Uniqueness

Unlike centralized systems that slow down as more robots join, each new AMR strengthens the FleetGrid network by extending range and redundancy — turning fleet scale from a liability into a competitive advantage.

4. Technical Approach & Methodology

Implementation proceeds through three cooperating levels, executed continuously and locally on every robot:

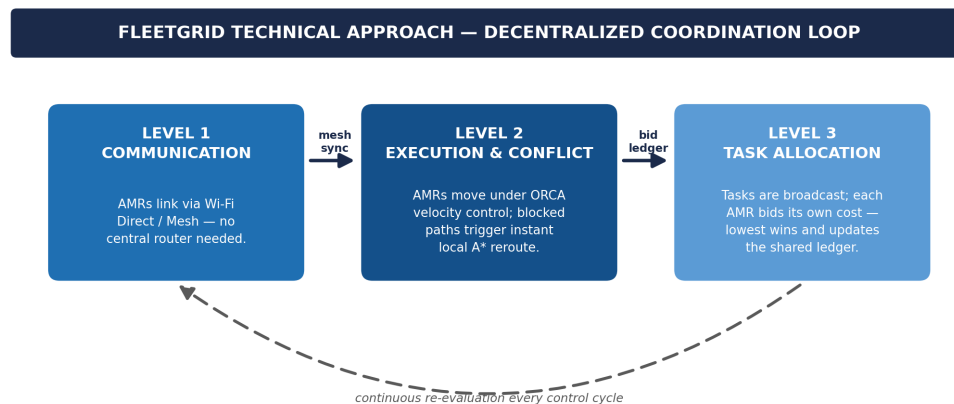


Figure 1: FleetGrid decentralized coordination loop — Communication, Execution & Conflict, and Task Allocation feed back into each other every control cycle

Level 1 — Communication

AMRs link via Wi-Fi Direct / mesh; no central router is needed.

Level 2 — Execution & Conflict

AMRs move under ORCA velocity control; when a path is blocked, the local obstacle map is updated and an instant local A* reroute is triggered.

Level 3 — Task Allocation

Tasks are broadcast to the fleet; each AMR calculates its own cost to complete the task, the lowest bid wins, and the shared ledger is updated accordingly.

5. Technology Stack

Layer	Technology
Hardware	NVIDIA Jetson Nano or Raspberry Pi 4 (edge compute) + 2D LiDAR (spatial awareness) + IMU (motion tracking)
Software & Algorithms	ROS 2 (Galactic / Humble) with eProsima Fast DDS for peer-to-peer communication (no ROS Master needed); ORCA for micro-collision avoidance; A* / DWA for local path planning
Communication	Wi-Fi Direct / Mesh via BATMAN-adv — no central router required
Dashboard & Control Layer	Node.js with Vue.js / React, served locally over a WebSocket listener on the edge network for real-time monitoring without external cloud servers

6. Literature Review & Research Foundation

FleetGrid's central design bet — that good fleet-wide behaviour can emerge from local sensing and local peer-to-peer decisions, without a central coordinator — is well supported by recent multi-robot coordination research. The clearest academic anchor for this approach is:

Li, Q., Gama, F., Ribeiro, A., & Prorok, A. (2020). *Graph Neural Networks for Decentralized Multi-Robot Path Planning*.

6.1 The Problem They Share with FleetGrid

The paper studies Multi-Agent Path Finding (MAPF): many robots, each with only a local field of view and a short-range communication radius, must reach their goals without collisions — with no single computer coordinating them. This is structurally the same challenge FleetGrid faces at warehouse choke points: robots that can only see and talk to nearby peers must still produce good fleet-wide, collision-free behaviour.

6.2 How the Paper Solves It

The authors train a neural pipeline — a CNN that compresses each robot's local view into a compact feature vector, a Graph Neural Network (GNN) that lets robots exchange and fuse those features with their communication-radius neighbours over K rounds of message passing, and an MLP that turns the fused result into a move — to imitate an optimal centralized planner (Conflict-Based Search) that is only run offline, during training. At run time each robot performs a cheap local forward pass (~ 1.9 ms) plus a few local message exchanges: no central computer is involved, which is exactly FleetGrid's operating model.

6.3 Why This Matters for FleetGrid

- Communication has measurable value: the paper shows that silent robots ($K=1$, no communication) degrade sharply as team size grows, while robots that exchange even one or two rounds of learned messages ($K=2$, $K=3$) sustain high success rates. This directly supports FleetGrid's bet on peer-to-peer telemetry sharing over MBIB and the mesh, rather than isolated local planning.

- Decentralized methods can outscale centralized ones: the classical centralized planner (CBS) becomes intractable beyond roughly 14 robots in dense maps, while the learned decentralized policy — trained on just 10 robots in a 20×20 warehouse-like grid — generalized to 60 robots in a 50×50 map with almost no performance loss (≥ 57 of 60 robots reaching goal in 97% of trials). This is strong independent evidence for FleetGrid's core claim that a decentralized architecture scales better than a central-server model as fleet size grows.
- Safety nets are still required: the paper's learned policy is not provably collision-free, so it relies on a simple “collision shielding” override (idle instead of colliding) — with the acknowledged risk of deadlocks. FleetGrid's own ORCA velocity control plus instant local A* reroute serves the same structural role, and the same deadlock risk applies (see Section 9).

6.4 A Natural Next Step for FleetGrid

MBIB currently resolves conflicts using a hand-crafted bidding rule (priority, battery, distance). The GNN paper's headline contribution is that this kind of local rule can instead be learned end-to-end by imitating an optimal planner offline. A promising future enhancement (see Section 12) is to replace or augment MBIB's fixed bidding weights with a lightweight learned GNN communication layer over FleetGrid's existing BATMAN-adv mesh — using the mesh's real neighbour graph as the GNN's adjacency matrix at each timestep.

7. Simulation Prototype — FleetGrid Digital Twin

The team has built a working simulation dashboard, FleetGrid, that models the proposed edge-AI distributed coordination system as a live digital twin of a warehouse floor (“AMR OPS — SIH”).

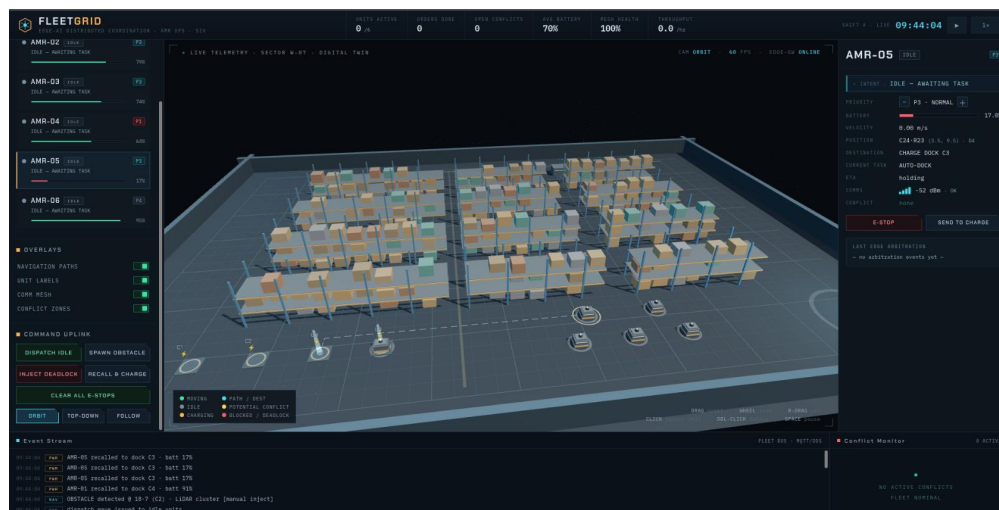


Figure 2: FleetGrid simulation dashboard — live 3D digital twin, fleet roster, and command uplink

Key elements visible in the prototype:

- Fleet roster (left panel): live status per AMR — idle / active, priority (P1–P4), and battery percentage, with AMR-05 shown selected.
- Live telemetry viewport (centre): a 3D digital twin of Sector W-07 showing warehouse racking, active units, charging docks, and navigation overlays (paths, unit labels, comm mesh, conflict zones), running at 60 FPS with the edge gateway marked ONLINE.

- Unit inspector (right panel): detailed state for the selected robot — intent, priority, battery, velocity, grid position, destination, current task, ETA, comms signal strength, and any active conflict — plus manual E-STOP and Send-to-Charge overrides.
- Command uplink (bottom-left): operator controls to dispatch idle units, spawn a test obstacle, inject a deadlock, recall-and-charge the fleet, or clear all E-stops — useful for demonstrating and stress-testing MBIB and collision handling live.
- Event stream and conflict monitor (bottom): a scrolling MQTT/DDS-style log of fleet-bus events (recalls, obstacle detections, dispatch waves) alongside a live count of active conflicts.
- Fleet-level KPIs (top bar): units active, orders done, open conflicts, average battery, mesh health, and system throughput (ops/hour), giving a judge-ready, at-a-glance view of fleet performance.

This prototype gives FleetGrid a concrete, demonstrable proof of concept for the SIH panel: it visualizes the mesh network, the per-robot decision state, and the fleet KPIs that the written proposal describes.

7.1 NEXUS-EDGE — Decentralized Mesh Consensus Build

A second build, NEXUS-EDGE (SIH-AMR-v2.4), demonstrates the P2P gossip consensus layer underneath FleetGrid in more detail — the mesh-level mechanics that make MBIB and task bidding possible without a central server.

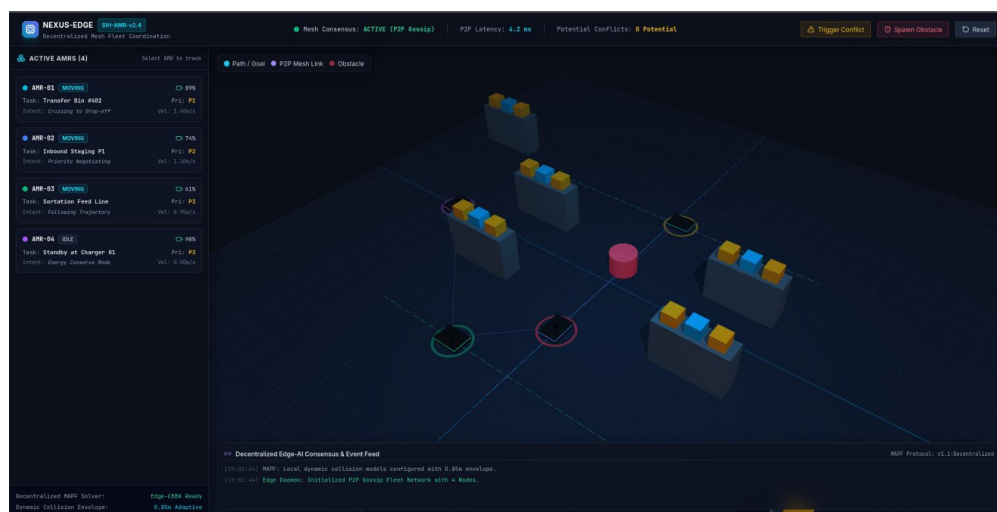


Figure 3: NEXUS-EDGE — decentralized mesh fleet coordination build, showing live P2P gossip consensus and per-unit intent

- Mesh consensus status bar: live confirmation that P2P Gossip consensus is ACTIVE, with measured P2P latency (4.2 ms) and a running count of potential conflicts — evidence that consensus is fast enough for real-time coordination.
- Active AMR roster: each unit's task, priority, and live intent in plain language (e.g. “Cruising to Drop-off,” “Priority Negotiating,” “Energy Conserve Mode”) alongside battery and velocity — making the MBIB bidding process observable rather than a black box.
- P2P mesh link overlay: the 3D view draws the live mesh topology (P2P mesh links) between robots and obstacles, distinct from planned paths/goals, so judges can see the communication graph the Section 6 literature review discusses — not just the physical motion.
- Operator test controls: “Trigger Conflict” and “Spawn Obstacle” let the team demonstrate MBIB conflict resolution and local A* re-routing live, on demand.
- Decentralized MAPF solver readout (bottom-left): confirms the active solver (Edge-CBBA) and the dynamic collision envelope (0.85 m, adaptive) — the same collision-shielding safety net discussed in Section 6.3.

Together, FleetGrid and NEXUS-EDGE give the team two complementary demo assets: FleetGrid communicates the full warehouse-scale picture (fleet KPIs, sector-wide digital twin), while NEXUS-EDGE zooms into the mesh-consensus mechanics that make the decentralized architecture credible to a technical judge.

8. Feasibility & Viability

8.1 Technical Feasibility

- Highly feasible using commercial off-the-shelf (COTS) hardware.
- ROS 2's native DDS already supports decentralized discovery, providing a proven software foundation.
- Running lightweight 2D mapping + ORCA on a Jetson Nano consumes under 40% CPU/GPU, leaving headroom for scaling.

8.2 Viability

- Industry-tested frameworks (ROS 2, DDS, ORCA) guarantee reliability in real deployments.
- Low compute overhead makes the system practical for continuous, 24/7 warehouse operation.
- Edge-based architecture reduces dependency on cloud infrastructure, improving resilience and efficiency.

8.3 Financial Viability

Metric	Value
Centralized annual infrastructure cost	₹15,00,000
Decentralized (FleetGrid) annual infrastructure cost	₹8,50,000
Annual saving	₹6.5 L
Percentage saving	43.3%

By removing warehouse-wide industrial Wi-Fi grids and central server racks, FleetGrid is projected to cut annual infrastructure cost by roughly 43%, while remaining deployable on COTS edge hardware.

9. Challenges & Mitigation Strategies

This section consolidates the risks flagged in the team's feasibility analysis with the operational failure modes documented in the decentralized-coordination literature (Section 6), since FleetGrid shares the same local-information, no-central-planner design.

Potential Challenge / Risk	Mitigation Strategy
Deadlocks at intersections — two robots each waiting for the other (a documented failure mode of local collision-avoidance policies)	MBIB priority bidding breaks ties deterministically (payload priority, battery, distance-to-goal); the dashboard's "Inject Deadlock" / "Recall & Charge" controls let operators detect and manually clear stuck units
Mesh communication delay or packet loss under real radio conditions (idealized instant comms assumed in simulation)	BATMAN-adv mesh multi-hop routing plus Spatial Broadcast Limiting keeps messages short-range and low-latency; ORCA velocity control degrades gracefully (slow/stop) rather than colliding if updates are stale

Potential Challenge / Risk	Mitigation Strategy
Mesh network partition (a group of robots temporarily isolated from the rest of the fleet)	Gossip protocol re-synchronizes the task ledger once connectivity is restored; each AMR continues executing its last-known task locally rather than freezing
Sensor / localization drift on low-cost LiDAR + IMU hardware	Local obstacle-map updates and periodic re-anchoring at charging docks (known fixed points) bound accumulated drift
Scaling validation — simulation results must still be proven on larger physical fleets	Phase rollout: validate on a small physical fleet first, then scale unit count in simulation and hardware in step, tracking mesh health and conflict-rate KPIs already built into the dashboard

10. Impact & Benefits

10.1 Impact

- Immediate Deployment: Tier-2/3 operators and legacy fulfillment centers can roll out autonomous swarms without waiting for costly infrastructure upgrades.
- Cost Savings: eliminates the need for warehouse-wide industrial Wi-Fi grids or server racks, slashing upfront investment.
- Plug-and-Play Scalability: enables smaller facilities to scale automation quickly, making advanced robotics accessible beyond top-tier warehouses.

10.2 Benefits

- Operational: continuous ORCA velocity-shifting keeps fleets moving, cutting task times by 25–30% compared to stop-and-wait.
- Economic: removes centralized cloud fees and heavy server maintenance, lowering ongoing costs.
- Environmental: localized edge computing consumes less power than constant cloud transmission, reducing the carbon footprint of warehouse robotics.

10.3 Alignment with UN Sustainable Development Goals

- SDG 8 — Decent Work and Economic Growth
- SDG 9 — Industry, Innovation and Infrastructure
- SDG 11 — Sustainable Cities and Communities
- SDG 12 — Responsible Consumption and Production

11. Business Model & Market Opportunity

This section builds on the team's feasibility and impact analysis (Sections 8 and 10) to frame FleetGrid as a commercial proposition, not just a technical one.

11.1 Target Market

The deck's own impact analysis points to a specific, underserved buyer: Tier-2/3 fulfillment operators and legacy warehouses that cannot justify the capital cost of a centralized fleet-management stack (industrial Wi-Fi grid,

server racks, licensed FMS software). FleetGrid's COTS-hardware, no-central-server design directly targets this segment — it is the difference between “automation for warehouses that can already afford a data centre” and “automation for warehouses that can't.”

11.2 Market Opportunity

Independent analyst estimates vary, but consistently point to a fast-growing addressable market:

- India's warehouse automation market is estimated at roughly ₹4,600–5,000 crore (~US\$0.56B) in 2025, projected to reach ~US\$1.5B by 2031 (~17.8% CAGR) — Mordor Intelligence.
- India's autonomous mobile robots (AMR) segment specifically is estimated at ~US\$150–315M in 2025, projected to grow at 15–19% CAGR through the early 2030s as e-commerce fulfillment and labour shortages push warehouses toward automation — IMARC Group / Grand View Research.

Even a modest share of the Tier-2/3 segment within this market represents a meaningful opportunity, and one that is systematically underserved by incumbent, server-dependent fleet-management platforms.

11.3 Key End-Users and Logistics Operators

India's logistics and retail majors are already investing in warehouse robotics — evidence that the buyer behaviour FleetGrid is targeting (automating fulfillment operations) is established and growing, even if today's deployments are concentrated among larger, well-capitalized players:

Company	Robotics Implementation
Flipkart & Myntra	Utilize advanced sorting and fulfillment robotics (via tech providers like GreyOrange) for high-speed e-commerce parcel processing.
Delhivery & DTDC	Deploy automated material handling and sorting solutions within large-scale logistics hubs.
Asian Paints	Uses automated guided and mobile robotic solutions for heavy-load movement across manufacturing and storage facilities.
Reliance Retail & Amazon India	Implement scalable AMR and AGV fleets across massive regional fulfillment centers to optimize goods-to-person picking.

These deployments validate demand at the top of the market; FleetGrid's opportunity is to bring the same category of automation to the Tier-2/3 operators and regional hubs below this tier, who cannot yet justify these players' centralized infrastructure spend.

11.4 Major AMR Providers & Manufacturers in India

FleetGrid's competitive landscape (Section 11.7) includes a growing set of Indian AMR hardware and platform vendors:

Provider	Core Offerings
Addverb	Provides heavy-duty AMRs like Dynamo to major manufacturing and warehouse facilities.
GreyOrange	Delivers AI-driven orchestration platforms and mobile robots via partnerships with distributors like Godrej Storage Solutions.
Ati Robotics (Ati Motors)	Builds the Sherpa line of heavy-duty AMRs and tuggers used in industrial and automotive warehouses.

Notably, these providers are largely hardware- and orchestration-platform-focused, typically paired with centralized fleet-management backends. FleetGrid does not compete on building robots — it is complementary, COTS-hardware-agnostic coordination software that could in principle run alongside AMRs from any of these vendors, while removing their dependence on centralized infrastructure.

11.5 Value Proposition

- ~43% lower annual infrastructure cost versus a centralized architecture (₹8.5L vs ₹15L/year, per Section 8.3) — a direct, quantified line-item saving a warehouse operator can put in a budget.
- No industrial Wi-Fi grid or server room required — shrinks the capital project from “IT infrastructure build” to “drop-in robot deployment.”
- 25–30% faster task completion versus stop-and-wait coordination (Section 10.2), improving fleet ROI independently of the infrastructure savings.
- Scale-positive network effect: each additional AMR extends mesh range and redundancy rather than adding load to a central server — the opposite of how centralized systems typically get more expensive and fragile as they grow.

11.6 Potential Revenue Model

As a Software-category PS submission, FleetGrid's commercial value sits primarily in the coordination software layer (MBIB, gossip protocol, mesh management, dashboard) rather than in hardware, which stays COTS and vendor-agnostic. A layered model follows naturally:

- Per-node coordination licence: an annual or subscription fee per AMR running the FleetGrid coordination stack — scales naturally with fleet size and with the “more robots = stronger network” uniqueness argument.
- Fleet Ops dashboard (SaaS): the Node.js/Vue.js monitoring and control layer (Section 5) as a hosted or on-prem subscription for warehouse operations teams.
- Deployment & integration services: one-time setup, LiDAR/IMU calibration, and warehouse-specific tuning for new customers — a natural services revenue line during onboarding.
- Support & mesh-health monitoring contracts: ongoing reliability assurance (uptime, conflict-rate SLAs) for operators running FleetGrid as mission-critical infrastructure.

11.7 Competitive Positioning

Incumbent warehouse fleet-management platforms — including the AMR providers listed in Section 11.4 — are largely built around a centralized architecture — a fleet-management server, dedicated industrial wireless infrastructure, and per-deployment customization. That architecture is defensible for large, well-capitalized distribution centres, but it prices out smaller operators and becomes a single point of failure at any scale. FleetGrid competes not by out-features these platforms, but by removing the infrastructure precondition altogether — positioning it as the accessible, self-scaling alternative for the long tail of Tier-2/3 warehouses that centralized vendors under-serve today.

11.8 Go-to-Market Approach

- Phase 1 — Pilot: deploy a small fleet (4–6 AMRs, matching the current simulation scale) in a single zone of a Tier-2/3 warehouse to validate cost savings and throughput gains against Section 8's projections.

- Phase 2 — Land and expand: leverage the network-effect uniqueness (Section 3.2) to grow fleet size within the same site — each additional robot improves mesh coverage, making expansion technically easier, not harder.
- Phase 3 — Horizontal scale: replicate the validated deployment playbook across additional Tier-2/3 sites and legacy fulfillment centres, using the per-node licensing model to align revenue growth directly with fleet growth.

12. Future Scope

- Learned communication (GNN-augmented MBIB): replace fixed bidding weights with a lightweight graph neural network layer over the existing BATMAN-adv mesh graph, trained offline by imitating an optimal centralized planner — following the approach validated by Li et al. (2020).
- Time-delayed / asynchronous GNN variants to explicitly model real mesh latency and packet loss, rather than assuming instant communication.
- Reinforcement-learning penalties on deadlock-inducing actions, to reduce reliance on the E-stop / recall safety net over time.
- Physical fleet trials to validate simulation KPIs (throughput, mesh health, conflict rate) against the FleetGrid dashboard's targets on real hardware.

13. Conclusion

FleetGrid proposes a serverless, edge-AI alternative to centralized AMR fleet management: robots sense locally, bid locally for right-of-way, and route messages through each other over a resilient mesh, removing the single point of failure and cost of a central server. The approach is grounded in current multi-robot coordination research — which shows that even simple local communication meaningfully improves decentralized coordination, and that such systems can scale well past the point where centralized planners become computationally intractable — and is backed by a working simulation prototype and a clear financial case ($\approx 43\%$ lower annual infrastructure cost). Together, these position FleetGrid as a practical, scalable path to bringing autonomous fleet coordination to warehouses of any size.

14. References

- Li, Q., Gama, F., Ribeiro, A., & Prorok, A. (2020). Graph Neural Networks for Decentralized Multi-Robot Path Planning.
- Team Vector6, Smart India Hackathon 2026 Idea Submission — “Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots in Smart Warehouses” (PS ID 26123).
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- Publicly reported robotics deployments — Flipkart, Myntra, Delhivery, DTDC, Asian Paints, Reliance Retail, Amazon India — and AMR provider offerings from Addverb, GreyOrange, and Ati Robotics (Ati Motors).